SECTION D: ENCRYPTION, KEYS & DIGITAL CERTIFICATES (10 MARKS)

Q8. Symmetric vs Asymmetric Encryption *[5 Marks]*

(a) Explain what is meant by the **Key Distribution Problem** in symmetric encryption: *[2]*

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(b) Complete the comparison table below: *[3]*

Feature	Symmetric Encryption	Asymmetric Encryption
Number of keys required per user pair		
Key used to decrypt message		
Relative processing speed (Fast / Slow)		

Q9. Asymmetric Key Exchange & Digital Certificates *[5 Marks]*

A student named David wants to purchase books from an online store securely.

(a) Which key will David's browser use to encrypt his credit card number? *[1]*
Answer: _____

(b) Which key will the online store use to decrypt David's credit card number? *[1]*
Answer: _____

(c) State TWO items contained within the store's SSL/TLS Digital Certificate that confirm its authenticity: *[2]*

Item 1: _____

Item 2: _____

(d) State the role of the Certificate Authority (CA): *[1]*

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